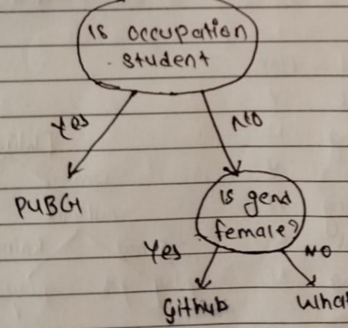
Decision tree:

* Example 1:

| Gender | Occupation | Suggestion |
|---|---|---|
| F | Student | PUBG |
| F | Programmer | GitHub |
| M | programmer | Whatsapp |
| F | Programmer | Github |
| M | Student | PuBG |
| M | Student | PUBG |

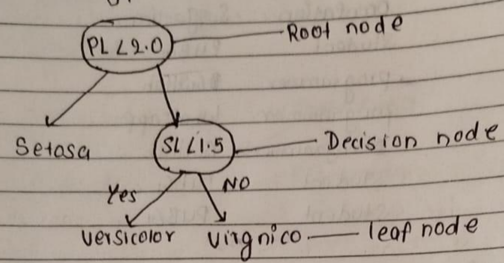
If occupation = student
    print (PUBG)
else
    If gender = Female
        Print (Github)
else
    Print (Whatsapp).

* What is Tree?




If occupation == student
    print (PUBG)
else
    If gender == female
        print (GithuB)
else
    print (what's app)

Terminology :



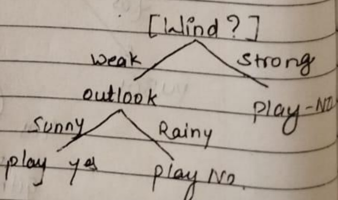

* Entropy :

Entropy measures how impure or uncertain a dataset is.

Ex1: out of Ice, water and vapor.
Entropy is high in vapor becouse of molecular free, random movement.

Ex2: More mixed → more randomness → higher entroply.
few mixed → more uniform → lower entropy.

| Student | outlook | Wind | Play |
|---|---|---|---|
| 1 | Sunny | weak | Y |
| 2 | Sunny | Strong | N |
| 3 | Sunny | weak | Y |
| 4 | Sunny | strong | N |
| 5 | Rainy | weak | N |
| 6 | Rainy | strong | N |




entropy $E(S) = \sum_{i=1}^{c} -P_i \log_2 P_i$

Date
Page

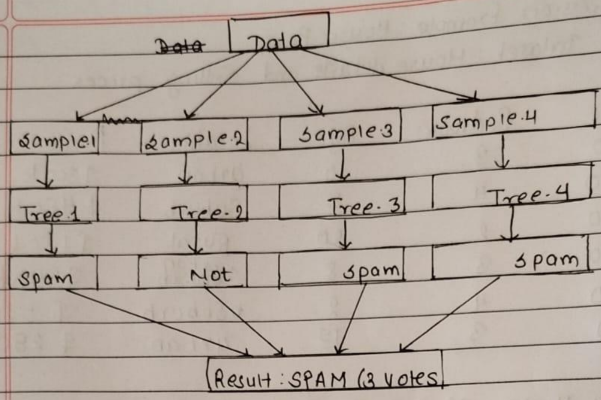
Information gain:

It measure how much uncertainty / impurety (entropy) decreases after splitting the data based on a feature.

Information gain = E(Parent) - [weighted Avg] * E(children)

A Decision Tree is a machine learning model that splits data into smaller subsets using condition on feature. It uses ~~the~~ entropy to measure impurity.

First the dataset is measured using entropy to determine how impure it is. Then the data is split based on different feature. Information Gain is used to measure how much entropy is reduced ~~each~~ after each split. The feature that gives the highest information Gain is selected as the best feature, and the data is split accordingly to achive better prediction.




Example (classification)

Dataset : email box :

| ID | Has "FREE" | Has "Win" | Many Links | Spam? |
|---|---|---|---|---|
| 1 | Yes | Yes | Yes | Spam Not |
| 2 | NO | NO | NO | SPam |
| 3 | Yes | NO | Yes | Spam Not |
| 4 | NO | NO | NO | Spam |
| 5 | Yes | Yes | NO | Spam Not |
| 6 | NO | NO | Yes | Spam |
| 7 | Yes | NO | NO | Spam Not |
| 8 | No | Yes | Yes | Spam |

New Email : Has "FREE", No "Win", Many links

~~Result : SPAM (5 out of~~

7 Trees vote :

Spam , spam , spam , not spam , spam , spam , not spam.

Result : SPAM (5 out of 7 votes)

* Random forest:
  → Many Decision Trees working together.

Think of it like:
- Asking 100 docters inserted of Just 1.
- Each docter give their opinion
- final answer = MaJority vote wins.

Advantages :-
- High accuracy : Better than Single tree
- No overfitting : Many trees fix errors.
- Both Type : Categories And Numbers
- find important features : knows What matters

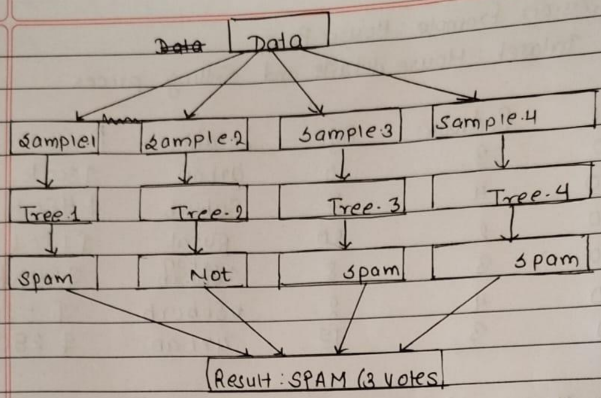
Real use:

Example :

| • categories prediction | Number Prediction |
|---|---|
| → Email: Spam or non Spam | → House price : $300.00 |
| → Disease : Sick or Healthy. | → Temperature : 25°C |
| → customer : will buy or not | |

- How Random forest works.

1. Take orginal Data
2. create Random Sample
3. Build Tree on ~~Each~~ Each Sample.
4. Each Tree votes
5. MaJority wins !

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Result : SPAM (5 out of 7 votes)

Regression Example : House Price
Dataset : House details and Selling prices.

| Size | Bedroom | Age | Location | Price |
|---|---|---|---|---|
| 1500 | 3 | 10 | urban | $300 k |
| 2000 | 4 | 5 | Suburb | $ 450 k |
| 1200 | 2 | 20 | Rural | $180 k |
| 1800 | 3 | 8 | ~~Suburb~~ urban | $350 k |
| 2500 | 4 | 2 | Suburb | $ 520 k |
| 1400 | 3 | 15 | urban | $ 280 . |

New House : Size = 1600, Beds = 3, Age = 12, urban.
Trees predict Different prices :
Tree 1 : $320 k   Tree 2 : 240k   Tree 3 : 295 k
Tree 4 : $ 350 k   Tree 5 : $ 315 k.

Avg price = $ 324,000

- Random Forest is a Bagging algorithm.
  ~~Bagg~~ : Bagging alg = BootStrap + Aggregating

- BootStrap = Random Sampling with replacement
- Aggregating = combine predictions from multiple models to produce one final result.